RESEARCH STATEMENT

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1. Introduction

My mathematical research is concerned primarily with finite groups and their character values. Specifically, I am interested in rationality properties of characters, including questions on how the structure of groups restricts the values of characters and vice versa.

A famous question of Richard Brauer, first introduced in [Br], asks the following: When is an arbitrary complex $n \times n$ matrix equal to the character table of some group. To date, there exists no set of necessary and sufficient conditions guaranteeing that such a matrix is a character table. Further, given an $n \times n$ matrix corresponding to some character table, we can ask how the entries of the matrix restrict the structure of the group. The majority of my work is motivated by these types of questions.

A complete list of my arXiv preprints may be found here.

2. Past Research

2.1. **Character-degree bounds for cyclotomic deficiencies.** My first publication (cf. [H1]) pertains to a conjecture and a question first posed by N. N. Hung and P. H. Tiep in [HT]. Before stating these, we introduce some terminology. We take $\mathbb{Q}_n$ to denote the extension of $\mathbb{Q}$ generated by a primitive nth root of unity. Given a cyclotomic integer α, we let $c(\alpha)$ denote the unique smallest integer such that $\alpha \in \mathbb{Q}_{c(\alpha)}$. Likewise, given a character χ of some group, $c(\chi)$ is defined to be the unique smallest integer such that all values of χ lie in $\mathbb{Q}_{c(\chi)}$. We also use $\mathbb{Q}(\chi)$ to denote the extension of $\mathbb{Q}$ generated by all values of χ. Here, $c(\alpha)$ and $c(\chi)$ are referred to as the *conductors* of α and χ respectively.

Hung and Tiep have posed the following conjecture.

Conjecture 1. ([HT, Conjecture 1.2]) *Let G be a group, and let $\chi \in \mathrm{Irr}(G)$. Then,*

$$|\mathbb{Q}_{c(\chi)} : \mathbb{Q}(\chi)| \leqslant \chi(1).$$

What this conjecture essentially asserts is that the field generated by an irreducible character is generally very close to being cyclotomic, and as such, the degree of the above field extension is sometimes called the *cyclotomic deficiency*. The conjecture

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is known to hold for numerous families of groups. From my joint work with Hung in [HH], I know that the conjecture also holds for the Suzuki 2-groups.

Hung and Tiep also posed a similar but purely number-theoretic question in their same paper:

Question 2. *Let $\alpha \in \mathbb{C}$ be expressible as a sum of n roots of unity. Is it true that*

$$|\mathbb{Q}_{c(\alpha)} : \mathbb{Q}(\alpha)| \leqslant n?$$

Using the theory of minimal vanishing sums (see [LL] or §3 of [PR]), I was able to show that the answer to the question is yes in the case that α is expressible as a sum of p-power roots of unity for some prime and in the case that α is expressible as a sum of two roots of unity. However, the above inequality seems to fail quite frequently. I wonder however whether the conjecture holds when we assume that α is expressible as a sum of roots of unity where each term has equal order since this special case would still have some interesting applications.

For the main result of [H1], I essentially bound the growth of counterexamples. In order to state the main result, we introduce the following function:

$$d(k) = \max\{|\mathbb{Q}_{c(\alpha)} : \mathbb{Q}(\alpha)| \;\big|\; \alpha \text{ is expressible as a sum of } k \text{ roots of unity.}\}.$$

The main result is as follows.

Theorem 3. *For $k \in \mathbb{Z}^+$, we have*

$$3^{\lfloor k/3 \rfloor} \leqslant d(k) < k! \left(\frac{3ke^{5k}}{\log(k+1)} \right)^k.$$

Although the upper bound may seem somewhat crude, the fact that the function $d(k)$ is finite for all k is somewhat surprising. The proof was made possible through the use of the theory of minimal vanishing sums but is otherwise combinatorial in nature. In my dissertation, I was also able to compute $d(3) = 4$ and $d(4) = 6$.

2.2. **Characters whose values are sums of few roots of unity.** In my more recent work in [H2], I have investigated characters whose values on nonidentity elements are sums of few roots of unity. This work generalizes previous work of Geoffrey Robsinson who in [R] investigated characters whose values on nonidentity elements were either roots of unity or zero. In [H2], I have essentially classified all possible characters satisfying this property. Below is an abbreviated version of my main result from that paper.

Theorem 4. *Let G be an abelian group having a generalized character χ whose values on nonidentity elements are sums of at most 2 roots of unity. Then, we have that $\chi = a\rho_G + \delta_1\lambda_1 + \delta_2\lambda_2$ for $a \in \mathbb{Z}$, $\delta_i \in \{-1, 0, 1\}$, and $\lambda_i \in \mathrm{Irr}(G)$, except possibly in the case where either 2, 3, or 5 divides $|G|$.*

Following in the footsteps of [R], I was able to use this result to obtain further results about the structure of nonabelian groups possessing such a character. As with my first paper, showing that these exceptional cases occur only when the group order is divisible by 2, 3, or 5 made use of the theory of minimal vanishing sums.

3. Current Work

3.1. **Feit's conjecture and related conjectures.** Most recently, I have been interested in a conjecture of Walter Feit and some related conjectures. The following conjecture was posed in [F]:

Conjecture 5. *Let G be a group, and choose $\chi \in \mathrm{Irr}(G)$. There exists $g \in G$ whose order equals $c(\chi)$.*

The conjecture is known to hold for solvable groups by the theory of π-special characters, and the conjecture has very recently been reduced to a more complicated conjecture regarding finite simple groups in [BKNT].

My current interest is in investigating stronger versions of Feit's conjecture for certain families of groups. In a recent collaboration with N. N. Hung, we initially sought to investigate the following question, which is a strengthening of Feit's conjecture:

Question 6. *Given a character $\chi \in \mathrm{Irr}(G)$ for a group G, under what circumstances does there exist $g \in G$ such that $c(\chi(g)) = c(\chi)$?*

The answer is no in general as there exists a supersolvable group G of order 960 possessing a counterexample. However, I know that the answer is yes when χ is primitive and G is solvable, using the theory of π-special characters. For nonsolvable groups, Hung and I have shown in [HH] that the answer is in fact yes for the groups $\mathrm{GL}(2,q)$ for q a prime power. For other families of groups, such as the Suzuki groups and $\mathrm{SL}(2,q)$, we were able to prove something yet stronger. In fact, if G is any such group, then for $\chi \in \mathrm{Irr}(G)$, there exists $g \in G$ for which $\mathbb{Q}(\chi(g)) = \mathbb{Q}(\chi)$. Gabriel Navarro has conjectured in [N, Conjecture 7.3] that this equality holds whenever χ has either odd degree or odd conductor, but this was recently disproven by Ulrich Thiel in [T]. However, none of the solvable counterexamples that Thiel has produced in this preprint are quasiprimitive. This motivates my most recent work [H4], where I proved the following:

Theorem 7. *Let G be a solvable group, and let $\chi \in \mathrm{Irr}(G)$ be a primitive (or even just fully factorizable) character such that either $\chi(1)$ is odd or χ is 2-rational. There exists $g \in G$ such that $\mathbb{Q}(\chi(g)) = \mathbb{Q}(\chi)$.*

3.2. **Classifying which p-groups are B-groups.** Given a group H, we say that H is a B-group if whenever G is a permutation group containing a regular subgroup isomorphic to H, then G is either imprimitive or doubly transitive. The study of B-groups was first initiated by William Burnside, who showed that any cyclic group

of composite prime-power order is a B-group. Since then, a major theme in the theory of permutation groups has been addressing whether a group satisfying certain properties is a B-group. In particular, work of C. H. Li in [Li] leads me to wonder the following:

Conjecture 8. *Let H be an abelian, noncyclic p-group for an odd prime p. Then, H is a B-group iff H cannot be decomposed as a direct product of subgroups each having equal cardinality.*

In [H3], using the O'Nan-Scott Theorem and the Classification of Finite Simple groups, I have classified completely the B-groups of order p^4 where p is any prime, and I have also shown the following:

Theorem 9. *Let H be a group of order p^q for primes p and q where $q < p$, and assume that $\exp(H) \geqslant p^2$. Then, H is a B-group.*

Another class of groups I am interested in investigating are the groups of square-free order. With the O'Nan-Scott Theorem in mind, a first step would be to classify the subgroups of square-free index in the finite simple groups.

Question 10. *Which groups of square-free order are B-groups?*

4. Future Projects

4.1. **Nonvanishing elements.** It has been shown by Burnside that any irreducible character with degree not equal to one has at least one conjugacy class on which the character evaluates to zero (cf. [Is, Theorem 3.15]). A sort of dual concept is the notion of a *nonvanishing class*. This is defined to be a conjugacy class on which none of the irreducible characters of G vanish. Likewise, we call a group element nonvanishing if it lies in a nonvanishing class. The study of such conjugacy classes was initiated in [INW]. In their paper, they introduced the following conjecture—an AI-generated proof of which was recently announced by D. Rossi in [Ro]:

Conjecture 11. *If G is a solvable group, then any nonvanishing class of G is contained in the Fitting subgroup $\mathbf{F}(G)$.*

The work surrounding this conjecture has led me to investigate the notion of nonvanishing elements in relation to a group's blocks. I have formulated the following conjecture:

Conjecture 12. *Let B be a p-block of a group G having full defect. If z lies in the center of some Sylow p-subgroup of G, then $\chi(z) \neq 0$ for any $\chi \in B$.*

It seems plausible to me that a better understanding of nonvanishing elements can be gained by restricting our attention to blocks. For instance, the only groups possessing a single p-block for any prime p are the Mathieu groups M_{22} and M_{24}, where $p = 2$ for both groups. At the same time, these are the only known finite

simple groups possessing nonvanishing elements, these elements lying in the centers of their Sylow 2-subgroups.

4.2. **Dade's Conjecture.** A more longstanding conjecture I find fascinating is one pertaining to so-called *monomial* characters, which arise from certain 1-dimensional representations of a group's subgroups. (See Ch. 5 of [Is] for a formal definition.) A group in which all irreducible characters are monomial is called an *M-group*. In [Do], Dornhoff showed that normal Hall subgroups of M-groups are themselves M-groups. He had also conjectured that, in fact, any normal subgroup of an M-group is again an M-group. Dade showed this to be false in [Da]. However, all known counterexamples to this conjecture have even order, so we have the following conjecture as a result:

Conjecture 13. *If G is an M-group of odd order, then every normal subgroup of G is also an M-group.*

4.3. **The number of minimal vanishing sums of a given weight up to a rotation.** We say that a sum of roots of unity is *minimal vanishing* if the sum is equal to zero and no proper subsum of the sum also equals zero. Minimal vanishing sums have consistently played a crucial role in my past research, and they appear to have a wide range of applications to the sciences. As such, they are interesting to study in their own right. The minimal vanishing sums having at most fourteen terms have been classified completely by several authors up to a rotation, i.e., multiplication through the terms by a root of unity. However, there exist no sharp, asymptotic results about the number of minimal vanishing sums up to a rotation. In [H2], I was able to derive a crude bound using a result of H. Mann in [Ma]. Namely, if $f(n)$ is the number of minimal vanishing sums up to a rotation having n terms, then $f(n) < e^{2n^2}$. I suspect that this bound can be improved drastically:

Question 14. *Can we produce an asymptotically tight bound on the number of minimal vanishing sums of roots of unity up to a rotation?*

4.4. **Existence of a Moore graph of valency 57.** Another problem I find highly fascinating is on the existence of a Moore graph of valency 57. While there are several equivalent definitions for a Moore graph, we can take a Moore graph to be a regular graph with diameter 2 whose smallest cycles have length 5. In an absolutely brilliant argument that considers the eigenvalues of a hypothetical Moore graph's adjacency matrix, one can show that such a graph must have valency 2, 3, 7, or 57. The Moore graphs of valency 2, 3, and 7 are respectively the 5-cycle, the Petersen graph, and the Hoffman-Singleton graph. However, the existence of a Moore graph of valency 57 is still unknown.

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